Enzo Luiz Fernandes

Electrical Engineer | Electronic hardware testing | C/C++ and embedded systems

LinkedIn · Technical portfolio

Profile

I am an electrical engineer with a degree from UNESP. I work with electronic hardware testing, component-level troubleshooting, bench instrumentation and laboratory operations. My academic and team projects include power electronics, PCB design and C/C++ firmware for ESP32.

Technical experience

Context	Tools and methods
Laboratory work	Oscilloscopes, multimeters, programmable supplies, hardware validation plans and fault investigation
Embedded projects	C/C++, ESP32, PlatformIO/VS Code, FreeRTOS, PWM, ADC and Bluetooth
Academic hardware project	Altium Designer, PCB design and LTspice simulation
Scripting and collaboration	Python for reports and equipment logs; Git/GitHub
Academic mathematics and control	MATLAB for plotting, signals, control simulation and applied mathematics

Professional experience

TDM GRID — Assistente Técnico de Proteção e Controle

September 2024 – Present

- I organize the technical routine of an electronics laboratory, including support, quality control and technical reports for Vebko test equipment and protection relays.
- I perform hardware tests and component-level troubleshooting using oscilloscopes, multimeters and programmable power supplies.
- I prepare and conduct hardware validation plans and operational checks, including IEC 61850 Sample Values features.
- I develop Python scripts to extract test-report data and process equipment logs, as well as routines related to service orders and inventory.
- I collaborate with Vebko on fault investigation and proposed hardware and software improvements.
- I participate in field commissioning and electrical tests, including activities in Ibirité-MG and at the GE factory in Taubaté.

Professional experience (continued)

TDM GRID — Estagiário de Proteção e Controle

January 2024 – September 2024

- I performed functional testing and maintenance of Vebko test equipment and protection relays, supporting fault identification and resolution.
- I supported customers and participated in GE protection relay configuration and testing.
- I carried out electrical protection tests using relay test equipment and checked equipment operation.

SQUADS — Engenharia de Garantia de Qualidade do Sistema (internship)

July 2020 – August 2021

- I implemented automated software tests with Cypress and BDD practices.
- I investigated failures and performed stress and volume tests, extracting and analyzing data to produce reports with Mochawesome.

Selected engineering projects

ESP32 BLDC controller — final degree project, UNESP

I developed and analyzed a controller prototype combining power-stage and PCB design in Altium, LTspice simulations and modular C/C++ firmware with PlatformIO/FreeRTOS, a hardware abstraction layer, ADC acquisition and serial/Wi-Fi telemetry. I carried out static bench tests and fault investigation, documenting the results in my published thesis.

Hardware, firmware and validation scope · Published thesis

TERA robotics — team projects

I contributed to ESP32/C++ firmware for locomotion, motor-stop commands on controller disconnection and weapon ESC control in CombatIO. My forks preserve the team history and link my individual commits. I also contributed to line-following and sumo robotics projects documented in the portfolio.

CombatIO contributions · Locomotiva contributions

Education

UNESP — Universidade Estadual Paulista "Júlio de Mesquita Filho", Ilha Solteira

Bachelor's degree in Electrical Engineering · February 2018 – August 2026

I completed my degree on **28 August 2026** and participated in the TERA robotics team.

Languages

Portuguese: native · English: fluent